

Benchmarking EFP2 Interaction Energies for Data-Driven Damping Function Optimization

INTRODUCTION

- Non-covalent interactions govern the structure, stability, and function of molecular and chemical systems.
- High-level quantum chemistry methods provide highly accurate interaction energies but are computationally expensive for large-scale applications.
- The Effective Fragment Potential (EFP) method offers a computationally efficient, quantum mechanically derived alternative for modeling intermolecular interactions.
- Symmetry Adapted Perturbation Theory (SAPT) and CCSD(T) extrapolated at the Complete Basis Set (CBS) interaction energies serve as the reference ("ground truth") and level of theory benchmark for evaluating the accuracy of EFP2 calculations.

OBJECTIVES

- Validate the accuracy of EFP2 using benchmark databases
- Compare EFP2 interaction energy with CBS reference values
- Provide data that will enable the optimization of EFP2 damping functions.

EFP ENERGY DECOMPOSITION

The total EFP interaction energy is decomposed into meaningful contributions.

$$E_{Total} = E_{Coul} + E_{Pol} + E_{Disp} + E_{ExRep} + E_{CT}$$

Have damping functions

Term	Physical Meaning
Electrostatic	Interaction between fragments multipoles
Polarization	Induction due to mutual electric fields
Dispersion	Interaction from electron fluctuations
Repulsion	Exchange repulsion from overlap of orbitals
Charge Transfer	Stabilization due to electron density

DAMPING FUNCTIONS

Damping functions attenuate interactions to avoid overestimation of energy terms.

Overlap-Based $\longrightarrow f_n^{TT}(i,j) = 1 - \left(\sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{(bR_{ij})^n}{n!} \right) e^{-bR_{ij}}$

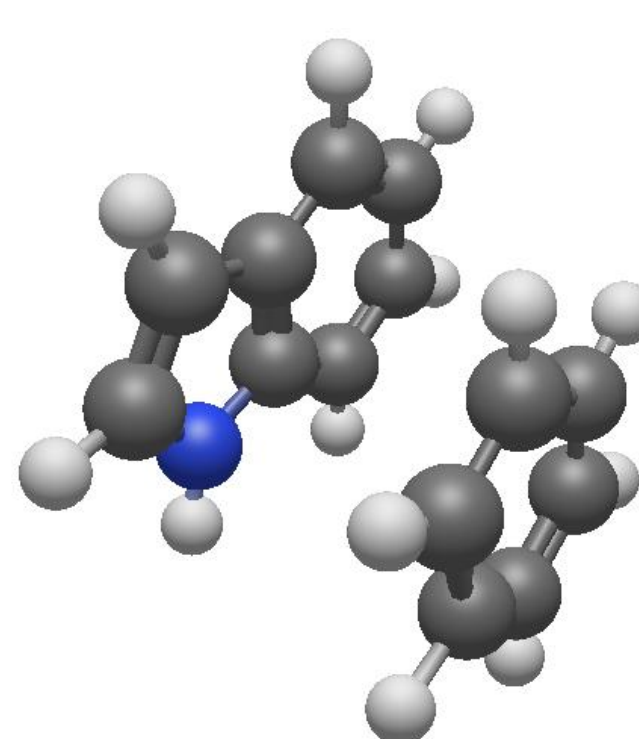
Tang-Toennies $\longrightarrow f_n^S(i,j) = 1 - S_{ij}^2 \sum_{n=0}^{6 \text{ or } 7 \text{ or } 8} \frac{-2\ln|S_{ij}|^{n/2}}{n!}$

BENCHMARK SYSTEMS

EFP2 performance is evaluated using benchmark databases

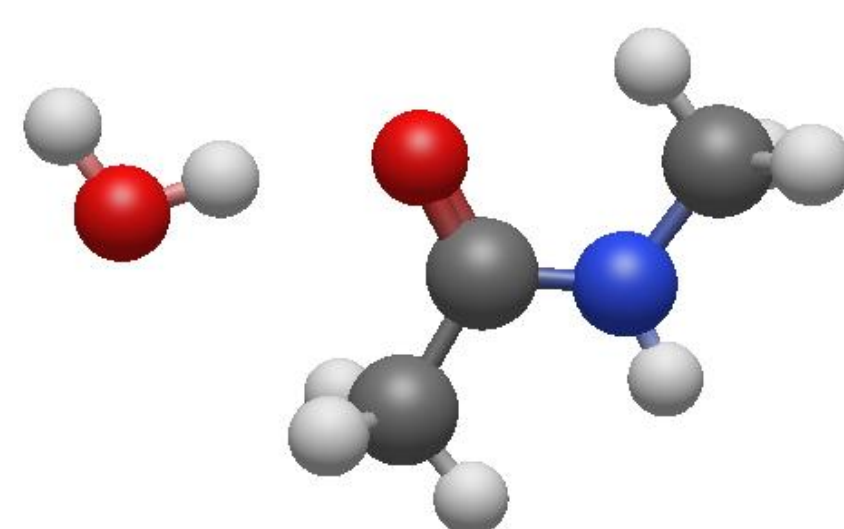
S22 Database

- 22 non-covalent dimers
- Characterized Interactions:
 - ✓ Hydrogen Bonds
 - ✓ Dispersion
 - ✓ Mixed Interactions



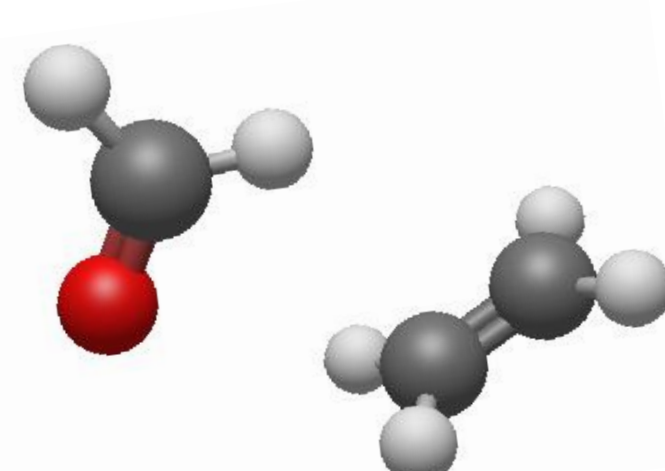
S66 Database

- 66 non-covalent dimers
- Broader variety of systems
- Includes:
 - ✓ Hydrogen Bonds
 - ✓ Dispersion
 - ✓ Mixed-Interactions



A24 Database

- 24 non-covalent dimers
- ✓ Hydrogen Bonds
- ✓ Dispersion Complexes
- ✓ Mixed-Interactions



ENERGY

Dispersion energy has three terms:

- Sums of the attractive forces to get the total dispersion energy

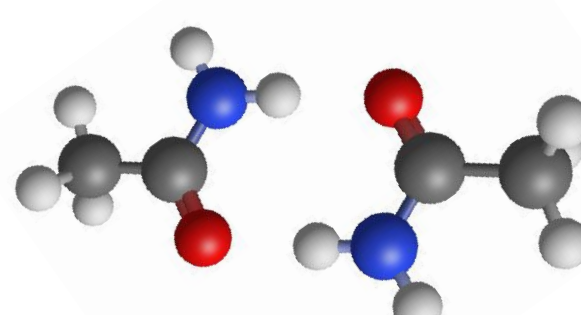
$$E^{Disp} = E(R^{-6}) + E(R^{-7}) + E(R^{-8}) = \sum_{n=6,7,8} E(R^{-n})$$

- Individual dispersion terms are calculated as the product of a damping function and its corresponding analytic expression.

$$E(R^{-n}) = f_n^{disp} * \text{analytic expression}$$

FRAGMENTS INTERACTION

Energy Component	EFP Energy (kcal/mol)
Electrostatic	-24.10
Repulsion	22.15
Polarization	-7.63
Dispersion	-6.37
Charge Transfer	-1.99
Final EFP Energy	-17.96

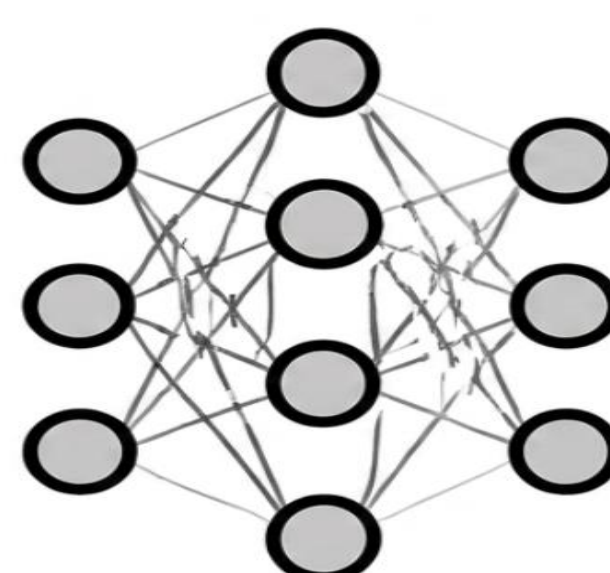


MACHINE LEARNING

- Machine learning is used to optimize the damping function parameters
- Provides a data-driven approach for parameter optimization

Input Features

- Overlap-based nonlinear transformations of 2-body terms



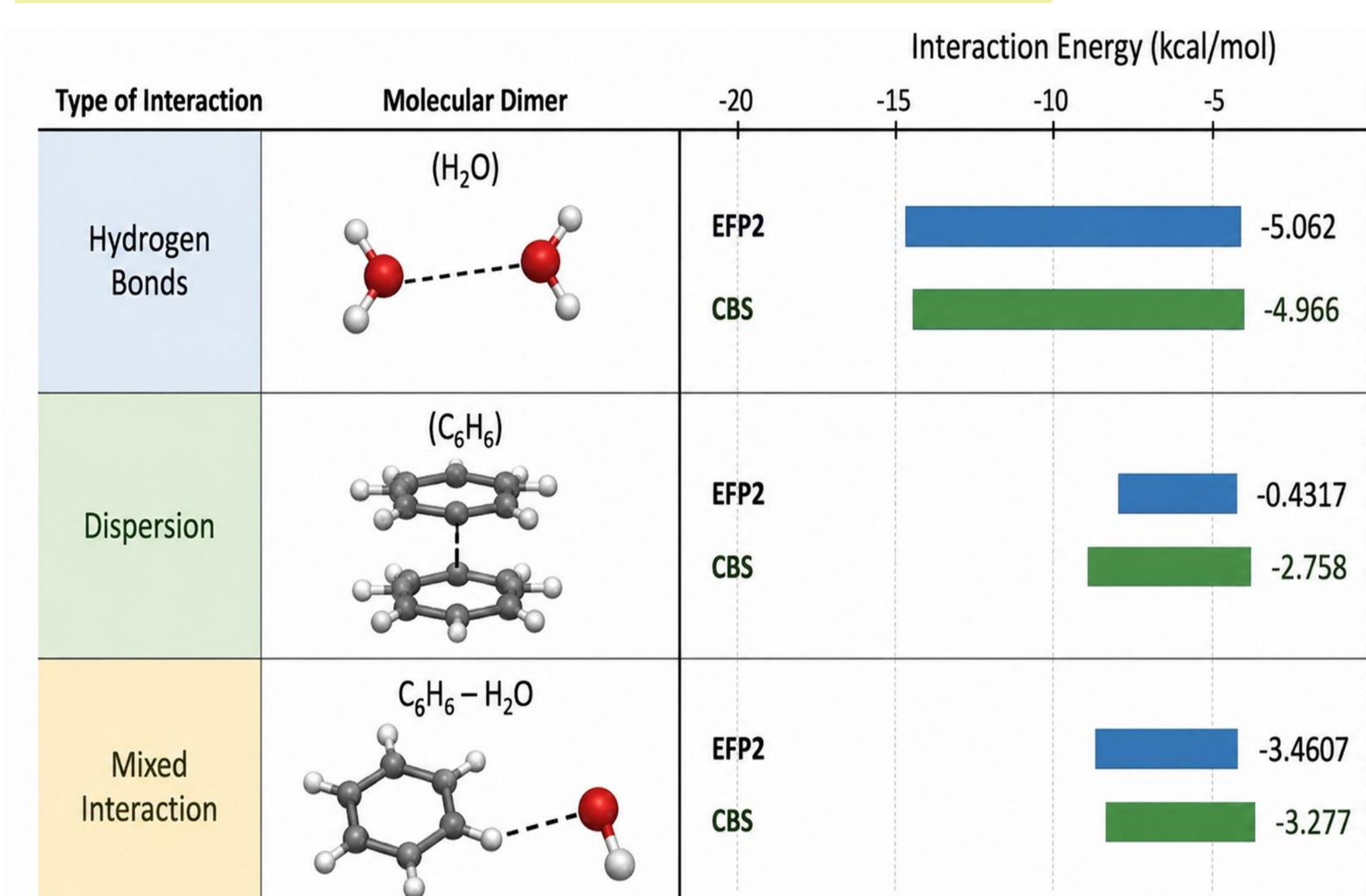
Model

- Multi-Layer Perceptron (MLP) Regressor neural network
- Iteratively optimizes features to get damped dispersion energy as close to SAPT benchmark dispersion energy as possible

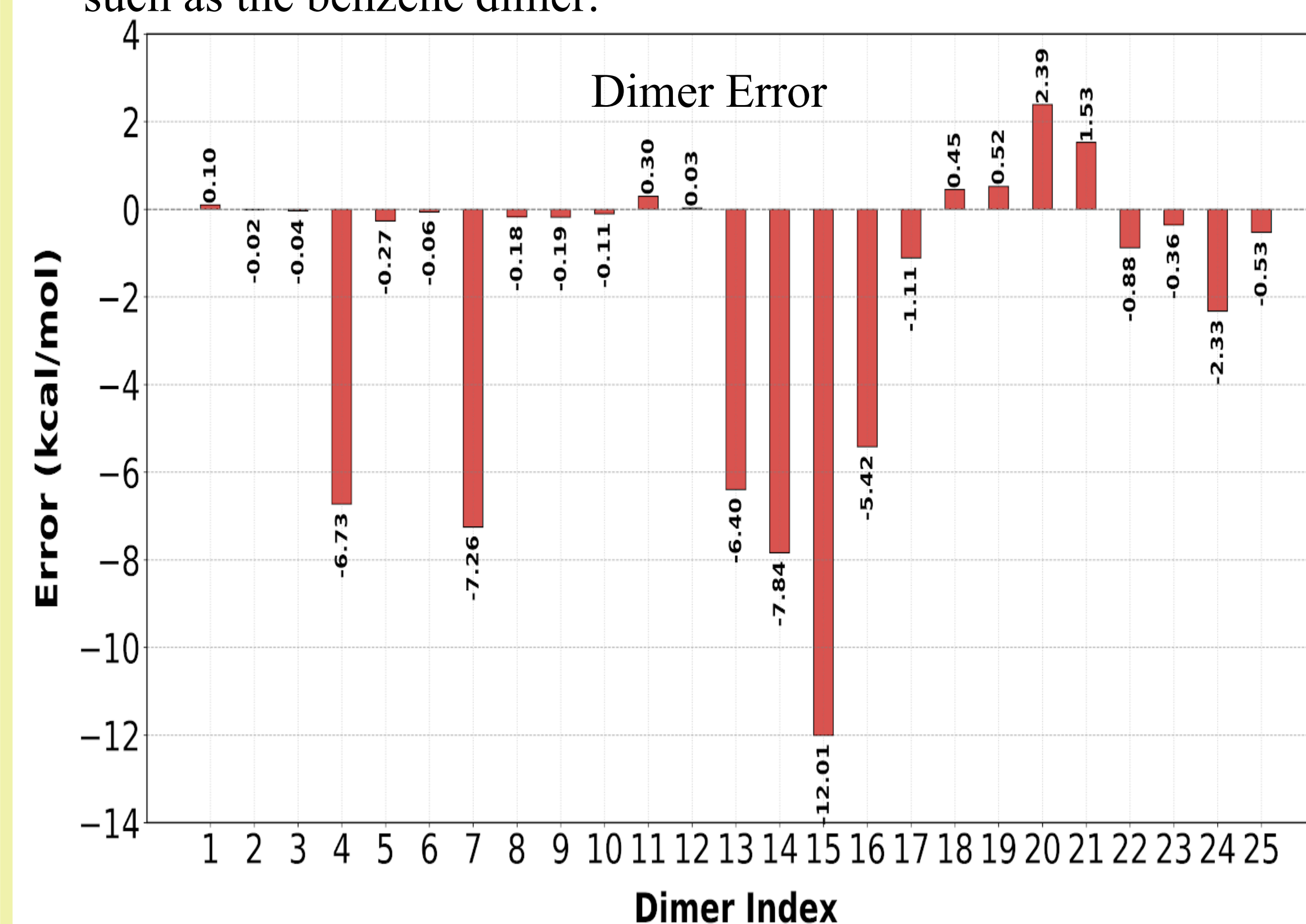
REFERENCES

- . *J. Chem. Theory Comput.* 2011, 7 (8), 2427–2438.
Phys. Chem. Chem. Phys. 2006; 8 (17): 1985–1993
Journal of Physical Chemistry A 105, 293–307 (2001).

RESULTS



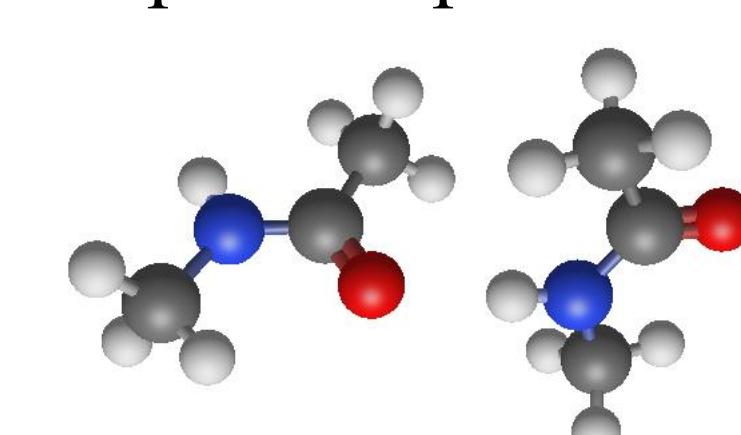
- EFP2 demonstrates high accuracy for electrostatic-dominated interactions, such as the water dimer, showing close agreement with CBS reference energies.
- Mixed interactions, like the benzene–water dimer, are also well-represented by EFP2, with results closely matching SAPT/CBS values.
- EFP2 performance decreases for dispersion-dominated systems, such as the benzene dimer.



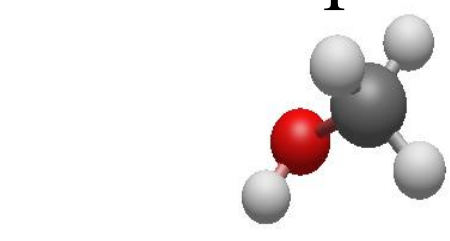
1	H ₂ O-H ₂ O
2	H ₂ O-MeOH
3	H ₂ O-MeNH ₂
4	H ₂ O-Peptide
5	MeOH-MeOH
6	MeOH-MeNH ₂
7	MeOH-Peptide
8	MeOH-H ₂ O
9	MeNH ₂ -MeOH
10	MeNH ₂ -MeNH ₂
11	MeNH ₂ -Peptide
12	MeNH ₂ -H ₂ O
13	Peptide-MeOH
14	Peptide-MeNH ₂
15	Peptide-Peptide
16	Peptide-H ₂ O
17	Uracil-Uracil
18	H ₂ O-Pyridine
19	MeOH-Pyridine
20	AcOH-AcOH
21	AcNH ₂ -AcNH ₂
22	AcOH-Uracil
23	AcNH ₂ -Uracil
24	Benzene-Benzene
25	Pyridine-Pyridine

- Increased orbital overlap enhances exchange repulsion, which EFP2 does not fully capture
- Small dimers show low errors because orbital overlap is minimal and electrostatic interactions dominate.
- EFP2 is not fully optimized for systems with strong orbital overlap

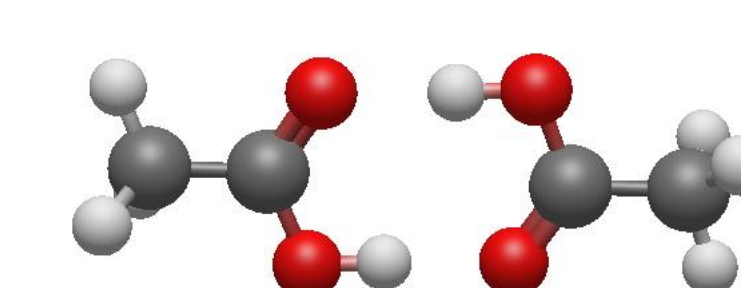
Peptide-Peptide



MeOH-Peptide



AcOH-AcOH



CONCLUSIONS

- EFP2 provides efficient interaction energies but errors increase for systems with significant orbital overlap.
- SAPT/CBS benchmarks provide accurate reference energies for EFP performance.
- Future work will focus on optimizing EFP2 damping functions using machine learning to better reproduce CBS reference interaction energies